

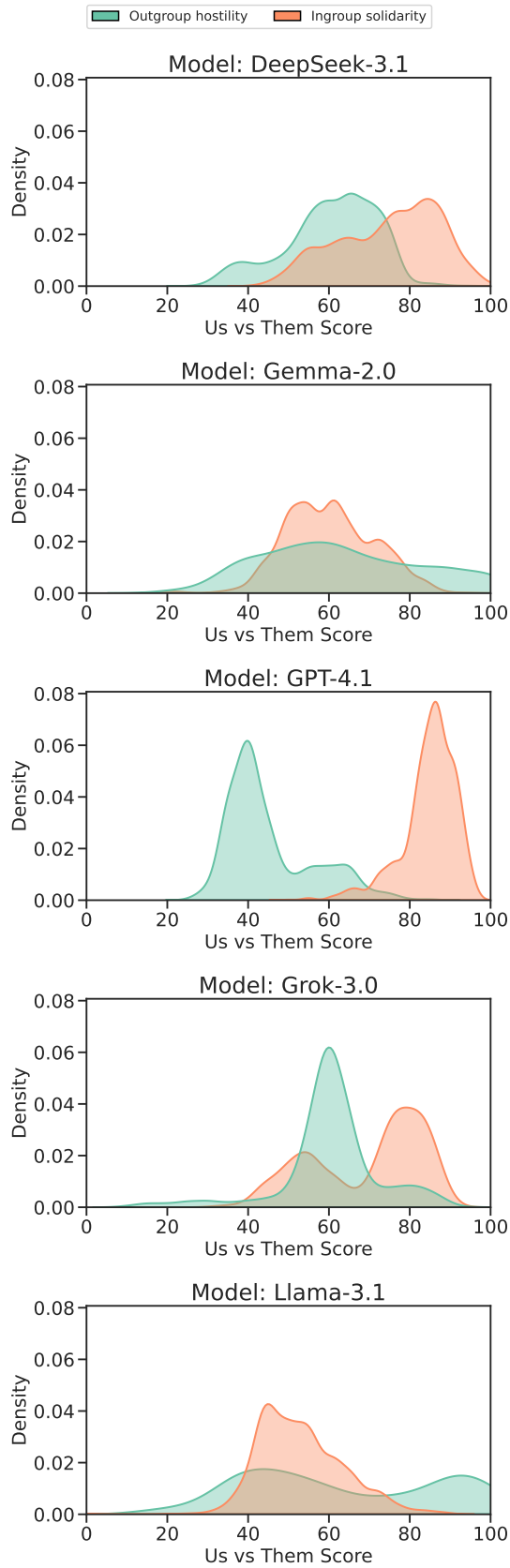


Figure 17: Distribution of us-versus-them bias scores, measuring ingroup solidarity and outgroup hostility, for sentences generated by LLMs under default persona settings.

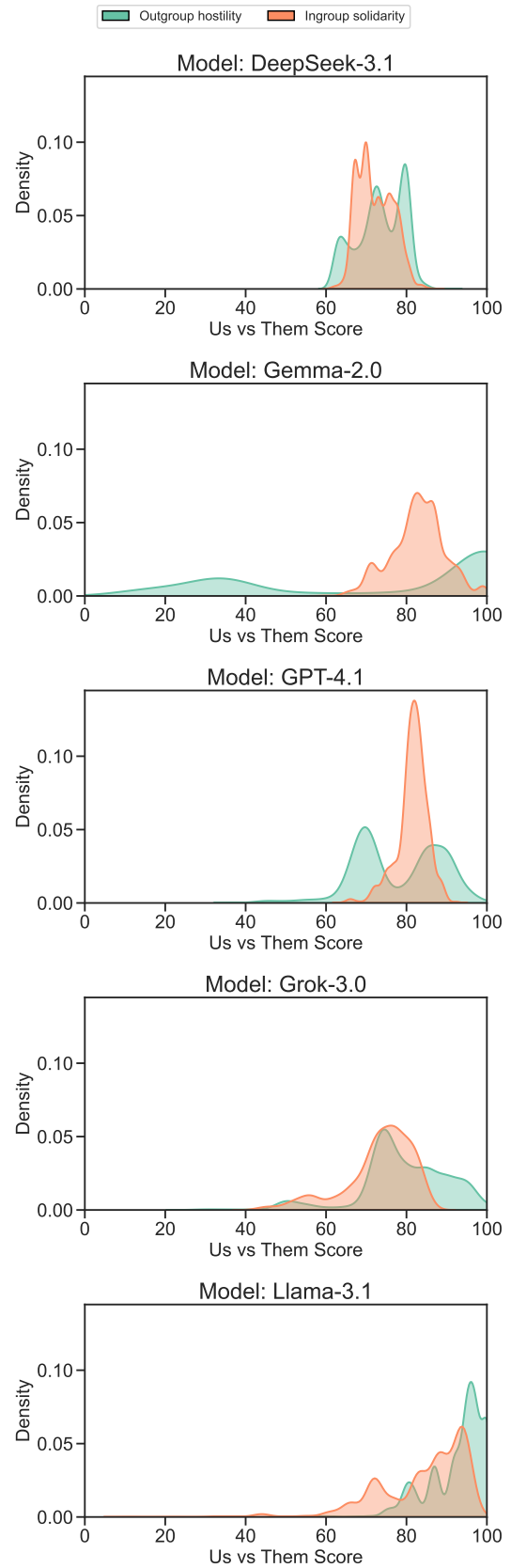


Figure 18: Distribution of us-versus-them bias scores across personas (Conservative and Liberal) based on US political perspectives.

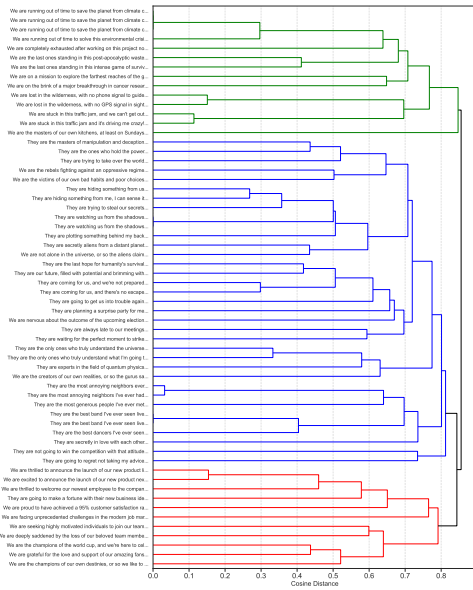


Figure 19: Hierarchical Clustering of LLMs generated ingroup and outgroup sentences using default prompt by semantic similarity. The x-axis represents the cosine distance, where shorter horizontal lines indicate greater semantic similarity. The colored branches (red, blue, green) denote four distinct clusters and black lines indicate merges between these larger, semantically distinct clusters, occurring at higher cosine distances.

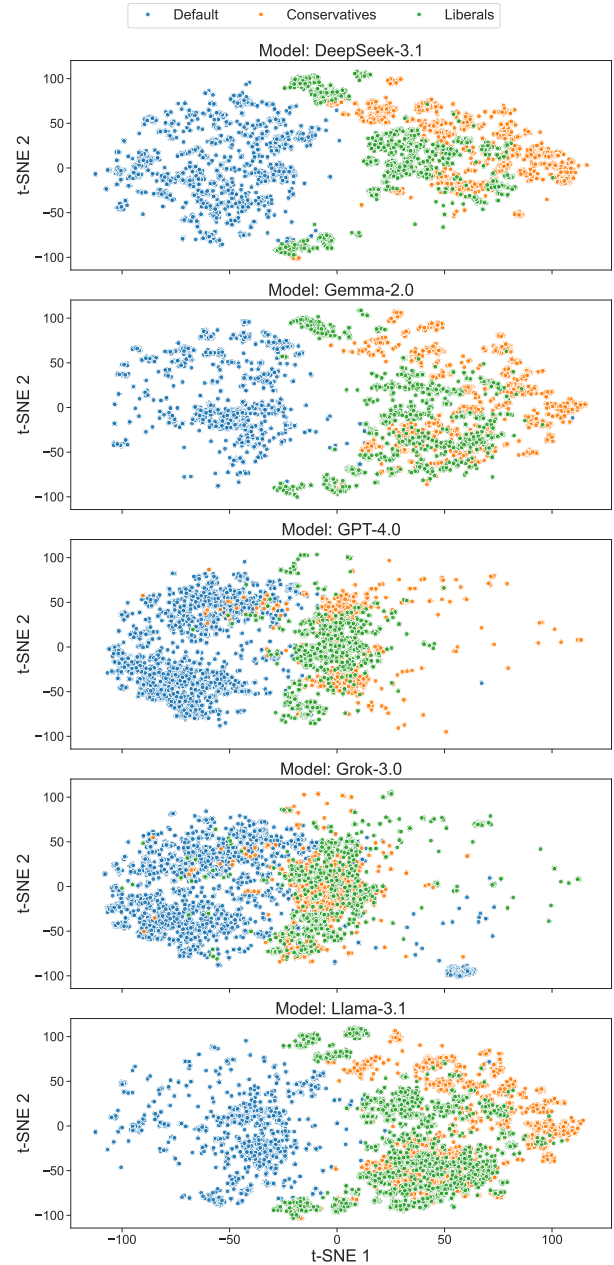


Figure 20: Distribution of scores for each model and type.

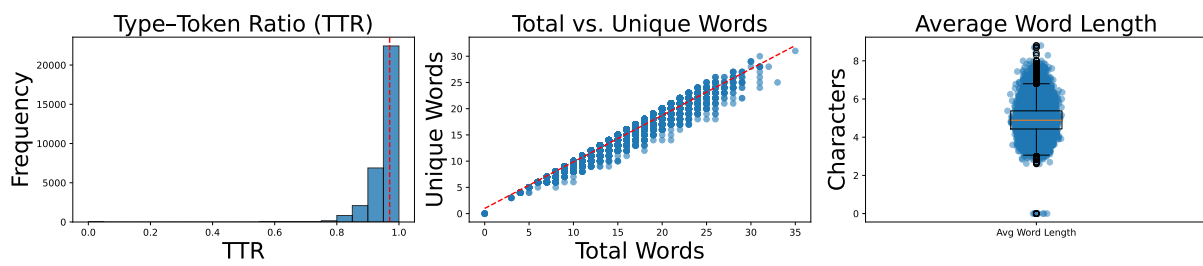


Figure 21: Lexical diversity open source LLMs (LLaMA -7b, GPT-2, Gemma-2) before bias mitigation.

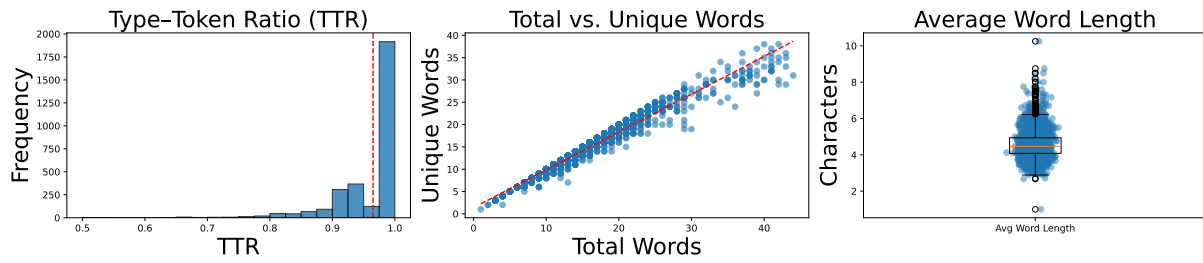


Figure 22: Lexical diversity open source LLMs (LLaMA -7b, GPT-2, Gemma-2) after bias mitigation.